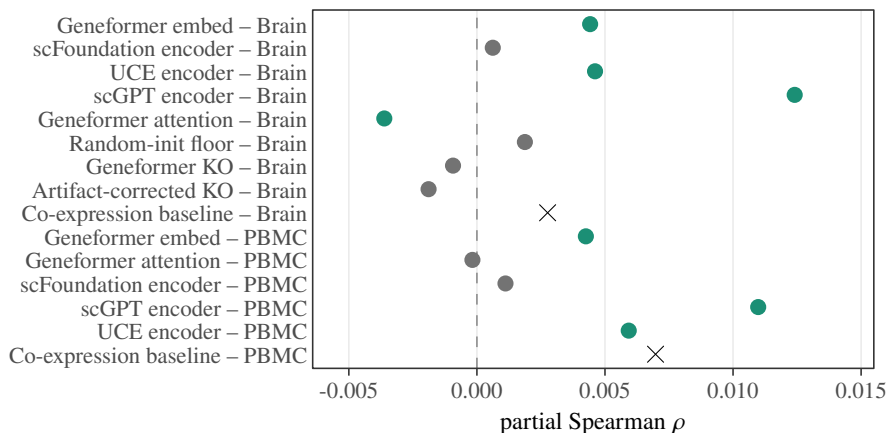


A

Full-confound specification (primary)

● co-expression baseline ● not supported ● supported by both nulls

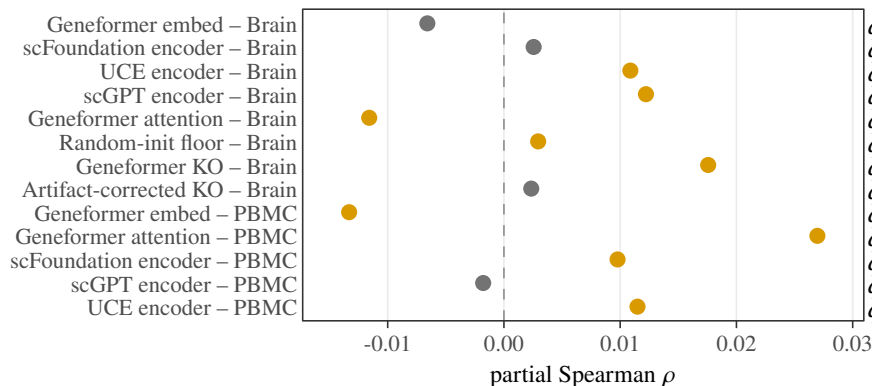


$q_M = 0.006$; $q_D = 0.003$
 $q_M = 0.633$; $q_D = 0.643$
 $q_M = 0.004$; $q_D = 0.003$
 $q_M = 0.004$; $q_D = 0.003$
 $q_M = 0.005$; $q_D = 0.016$
 $q_M = 0.243$; $q_D = 0.233$
 $q_M = 0.465$; $q_D = 0.418$
 $q_M = 0.160$; $q_D = 0.146$
 $p_M = 0.019$; $p_D = 0.014$
 $q_M = 0.002$; $q_D = 0.003$
 $q_M = 0.866$; $q_D = 0.839$
 $q_M = 0.491$; $q_D = 0.449$
 $q_M = 0.002$; $q_D = 0.003$
 $q_M = 0.002$; $q_D = 0.003$
 $p_M = 0.001$; $p_D = 0.001$

B

Non-degree co-primary (closest q shown; none below 0.05)

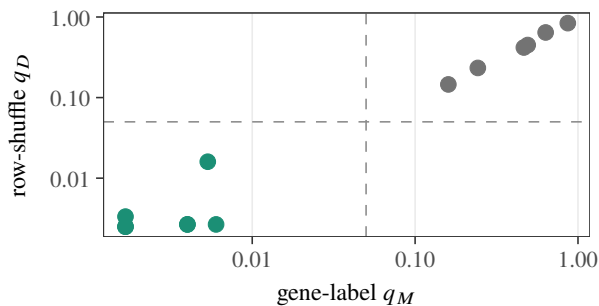
● not supported ● supported by one null



$q_M = 0.593$; $q_D = 0.689$
 $q_M = 0.852$; $q_D = 0.897$
 $q_M = 0.354$; $q_D = 0.003$
 $q_M = 0.354$; $q_D = 0.018$
 $q_M = 0.593$; $q_D = 0.003$
 $q_M = 0.354$; $q_D = 0.045$
 $q_M = 0.354$; $q_D = 0.003$
 $q_M = 0.852$; $q_D = 0.897$
 $q_M = 0.328$; $q_D = 0.002$
 $q_M = 0.328$; $q_D = 0.002$
 $q_M = 0.338$; $q_D = 0.014$
 $q_M = 0.856$; $q_D = 0.998$
 $q_M = 0.328$; $q_D = 0.002$

C

Both nulls below 0.05



D

Dual-null rows vs baseline ($r^2 = \rho^2$ on FM bars)